SCIENTIFIC MACHINE LEARNING PSET 01.

PROBLEM 1.1

(a) Note that if $w^T x^{(i)} + b > 0$, the correct label would be $y^{(i)} = 1 > 0$ so $\hat{\gamma}^{(i)} = y^{(i)}(w^T x^{(i)} + b) > 0$. Similarly, if $w^T x^{(i)} + b < 0$, the correct label would be $y^{(i)} = -1 < 0$ so $\hat{\gamma}^{(i)} = y^{(i)}(w^T x^{(i)} + b) > 0$.

(b) Suppose that $\hat{\gamma}^{(i)} > 0$ for (w, b). Then $\tilde{\hat{\gamma}}^{(i)} = y^{(i)}(2w^T x^{(i)} + 2b) = 2y^{(i)}(w^T x^{(i)} + b) = 2\hat{\gamma}^{(i)} > 0$. By part (a), our new predictions are also correct.

(c) Let $\vec{x_0}$ be a point on the decision boundary. Then, plugging into $h(\vec{x}) \equiv \vec{w}^T \vec{x} + b = 0$, we find $b = -\vec{w}^T \vec{x_0}$. So, $h(\vec{x}) = \vec{w}^T(\vec{x} - \vec{x_0})$.

Note that $\vec{x} - \vec{x_0}$ is the vector along the hyperplane $\forall \vec{x}$ points on the hyperplane i.e., for all $\vec{x}$ that satisfy $\vec{w}^T(\vec{x} - \vec{x_0}) = 0$. So, $\vec{w}$ is orthogonal to the vector along the hyperplane.

(d)

If A is on the right of DB, i.e., $y^{(i)} > 0$: $\vec{x_B} + \vec{x}_{BA} = \vec{x_A}$

otherwise i.e., $y^{(i)} < 0$: $\vec{x_B} - \vec{x}_{BA} = \vec{x_A}$

So: $\vec{x}_B = \vec{x_A} - y^{(i)} \vec{x}_{BA}$ but $\vec{x}_{BA} = \gamma^{(i)} \dfrac{\vec{w}}{\|w\|}$

$$\vec{x_B} = \vec{x}^{(i)} - y^{(i)} \gamma^{(i)} \frac{\vec{w}}{\|w\|}$$

(e) $\vec{w}^T \vec{x_B} + b = 0$

$$\vec{w}^T \left(\vec{x}^{(i)} - y^{(i)} \gamma^{(i)} \frac{\vec{w}}{\|w\|} \right) + b = 0$$

$$\vec{w}^T \vec{x}^{(i)} - y^{(i)} \gamma^{(i)} \|w\| + b = 0$$

$$\vec{w}^T \vec{x}^{(i)} + b = y^{(i)} \gamma^{(i)} \|w\|$$

$$\gamma^{(i)} = \frac{1}{y^{(i)}} \left(\left(\frac{w}{\|w\|} \right)^T \vec{x}^{(i)} + \frac{b}{\|w\|} \right)$$

but $y^{(i)} = 1, -1$ so $\dfrac{1}{y^{(i)}} = y^{(i)}$

$$\Rightarrow \gamma^{(i)} = y^{(i)} \left(\left(\frac{w}{\|w\|} \right)^T x^{(i)} + \frac{b}{\|w\|} \right)$$

PROBLEM 1.2.

(a) Suppose $g_i(w) \leq 0$ for all w. Then $\mathcal{L}(w,\alpha) = f(w) + \sum_{i=1}^{k} \alpha_i g_i(w)$ is maximized upon setting all $\alpha_i = 0$. So, $\theta_P(w) = f(w)$ in that case.

Now, suppose $\exists\, 1 \leq j \leq k$ s.t. $g_j(w) > 0$. Then, $\mathcal{L}(w,\alpha)$ is maximized when we set $\alpha_j \to \infty$ and in this case $\theta_P(w) = \infty$.

(b) $d^* = \max_{\alpha \geq 0} \min_{w} \mathcal{L}(w,\alpha)$, $\quad p^* = \min_{w} \max_{\alpha \geq 0} \mathcal{L}(w,\alpha)$

Let $\bar{w}$ be s.t. $\mathcal{L}(\bar{w},\alpha) = \min_{w}(\mathcal{L},\alpha)$. Then

$$\mathcal{L}(\bar{w},\alpha) \leq \mathcal{L}(w,\alpha) \quad \forall w,\alpha.$$

$$\max_{\alpha} \mathcal{L}(\bar{w},\alpha) \leq \max_{\alpha} \mathcal{L}(w,\alpha) \quad \forall w$$

Then in particular, $\max_{\alpha} \mathcal{L}(\bar{w},\alpha) \leq \min_{w} \max_{\alpha} \mathcal{L}(w,\alpha)$

$$\Rightarrow \quad d^* \leq p^* \ \square$$

(c) If $\alpha_i^* g_i(w^*) = 0$ and $\alpha_i > 0$, then $g_i(w^*) = 0$. This corresponds to the datapoints which saturate the bound. In our case, this corresponds to the datapoints which minimize distance with the decision boundary, i.e., this corresponds to the support vectors.

$$\left[\, g^i(w^*) = 0 \iff y^{(i)}(w^T x^{(i)} + b) = \gamma = \min_i \gamma^{(i)} \,\right]$$

PROBLEM 1.3

(a) Want $y^{(i)}(w^T x^{(i)} + b) \geq 1$

$$\Rightarrow 1 - y^{(i)}(w^T x^{(i)} + b) \leq 0$$

Take $g_i(w) = 1 - y^{(i)}(w^T x^{(i)} + b)$

(b) $$\mathcal{L}(w, b, \alpha) = \frac{1}{2}\|w\|^2 + \sum_{i=1}^{n} \alpha_i g_i(w)$$

$$= \frac{1}{2}\|w\|^2 + \sum_{i=1}^{n} \alpha_i (1 - y^{(i)}(w^T x^{(i)} + b))$$

(c) $$\nabla_w \mathcal{L}(w, b, \alpha) = w^* + \sum_{i=1}^{n} \alpha_i (-y^{(i)} x^{(i)}) = 0 \Rightarrow w^* = \sum_{i=1}^{n} \alpha_i y^{(i)} x^{(i)}$$

$$\frac{\partial \mathcal{L}}{\partial b}(w, b, \alpha) = -\sum_{i=1}^{n} \alpha_i y^{(i)} = 0 \Rightarrow \sum_{i=1}^{n} \alpha_i y^{(i)} = 0$$

(d) $$\|w\|^2 = w^T w = \sum_{i,j=1}^{n} \alpha_i y^{(i)} (x^{(i)})^T \alpha_j y^{(j)} x^{(j)}$$

$$\sum_{i=1}^{n} \alpha_i (y^{(i)} w^T x^{(i)} + y^{(i)} b) = \sum_{i,j=1}^{n} \alpha_i y^{(i)} \alpha_j y^{(j)} (x^{(j)})^T x^{(i)} + \cancel{b \sum_{i=1}^{n} \alpha_i y^{(i)}}^{\,0}$$

$$\overset{i \leftrightarrow j}{=} \sum_{i,j=1}^{n} \alpha_i \alpha_j y^{(i)} y^{(j)} (x^{(i)})^T x^{(j)}$$

Then,

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} \sum_{i,j=1}^{n} \alpha_i y^{(i)} (x^{(i)})^T \alpha_j y^{(j)} x^{(j)} - \sum_{i,j=1}^{n} \alpha_i y^{(i)} (x^{(i)})^T \alpha_j y^{(j)} x^{(j)} + \sum_{i=1}^{n} \alpha_i$$

$$= \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{n} \alpha_i y^{(i)} (x^{(i)})^T \alpha_j y^{(j)} x^{(j)}$$

(e) We already have the slope w^* and want to pick the intercept that maximizes the functional margin. Since the functional margin is the distance from the closest point to the DB, the DB cannot be closer to the closest point with $y^{(i)} = 1$ or $y^{(i)} = -1$, i.e., it must be in the center of both the points:

(e) We want to minimize $\mathcal{L}(w,b,\alpha) = \frac{1}{2}\|w\|^2 + \sum_{i=1}^{n} \alpha_i(1 - y^{(i)}(w^T x^{(i)} + b))$

So, we could maximize $\sum_{i=1}^{n} \alpha_i(1 - y^{(i)}(w^T x^{(i)} + b))$. Since $\alpha_i \geq 0$, we could instead minimize $y^{(i)}(w^T x^{(i)} + b)$ $\forall i$. Let w^* be the optimal value. Then, we want to minimize $y^{(i)}(w^{*T} x^{(i)} + b)$ for all i.

If $y_i = 1$, we want $\min (w^*)^T x^i + b \rightarrow$ Choose $b = -\min_i (w^*)^T x^i$.

$$\gamma^{(i)} = y^{(i)}((w^*)^T x^{(i)} + b)$$

Take points + and - s.t. $y_+ = 1$ and it is the closest (to DB) such point and $y_- = -1$ and it is the closest (to DB) such point.

Then:

$$\gamma^{(+)} - b^* = w^{*T} x^{(+)}$$
$$\gamma^{(-)} + b^* = -w^{*T} x^{(-)}$$

$$\Rightarrow \underbrace{\cancel{\gamma^{(-)} - \gamma^{(+)}}}_{0 \text{ since } \gamma^{(+)} = \gamma^{(-)}} + 2b^* = -w^{*T} x^{(-)} - w^{*T} x^{(+)}$$

$$b^* = -\frac{1}{2}\left[\min_{i:y^{(i)}=1} w^{*T} x^{(i)} + \max_{i:y^{(i)}=-1} w^{*T} x^{(i)}\right]$$

The last step follows because if $y^{(i)} = 1$, then $\gamma^{(i)} = w^T x^{(i)} + b$. We are trying to minimize $\gamma^{(i)}$ so we would like to minimize $w^T x^{(i)}$:

$$\min_{i:y^{(i)}=1} \gamma^{(i)} = \min_{i:y^{(i)}=1} [w^T x^{(i)} + b^*] = \min_{i:y^{(i)}=1} w^T x^{(i)} + b^*$$

$$\Rightarrow b^* = \min_{i:y^{(i)}=1} \gamma^{(i)} - \min_{i:y^{(i)}=1} w^T x^{(i)}$$

But $\min_{i:y^{(i)}=1} \gamma^{(i)} = \gamma^{(+)}$ so $w^{*T} x^{(+)} = \min_{i:y^{(i)}=1} w^T x^{(i)}$

Similarly, if $y^{(i)} = -1$ then $\gamma^{(i)} = -(w^T x^{(i)} + b)$. So, we are trying to maximize $w^T x^{(i)}$ to minimize $\gamma^{(i)}$:

$$\min_{i:y^{(i)}=-1} \gamma^{(i)} = -\max_{i:y^{(i)}=-1} [w^T x^{(i)} + b^*] = -\max_{i:y^{(i)}=-1} w^T x^{(i)} - b^*$$

$$\Rightarrow b^* = -\min_{i:y^{(i)}=-1} \gamma^{(i)} - \max_{i:y^{(i)}=-1} w^T x^{(i)}$$
$$= -\gamma^- - \max_{i:y^{(i)}=-1} w^T x^{(i)}$$

so $w^{*T} x^{(-)} = \max_{i:y^{(i)}=-1} w^T x^{(i)}$.

(f) $w^{*T} x + b^* = \left(\sum_{i=1}^{n} \alpha_i y^{(i)} x^{(i)}\right)^T x + b^*$

From 1.2, recall that if $g_i(w_k) < 0$ then $\alpha_i = 0$ and $\alpha_i > 0$ if $g_i(w^*) = 0$ (i.e., support vectors)

so we can get rid of datapoints in the sum which are not support vectors.

$$w^{*T}x + b^* = \sum_{i:\alpha_i > 0} \alpha_i y^{(i)} x^{(i)T} x + b^*$$

$$= \sum_{i:\alpha_i > 0} \alpha_i y^{(i)} \langle x^{(i)}, x \rangle + b^* \quad \square$$

PROBLEM 2.

(a) on code

(b) Our model performs well at low temperatures and seems to be getting better at higher temperatures (but still bad) after hitting its lowest around 2.5K. This is around the critical temperature of the model which is what we would expect. Since we have phase transitions / large jump near T_c, the model becomes more uncertain.

We could try to use a physics-informed approach and use e.g. magnetization as a feature.

(d) The SVC works much better than logistic regression. I used polynomial and rbf kernels to encode the correlations better than the logistic regression. There is still a dip near T_c but it is narrower and makes sense since there is no defined phase there. But, the non-linear kernels allow more freedom in the decision boundaries and improve the model.

Collaborators.

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